

BUSINESS REQUIREMENTS

Customer Dashboard Devices & Hydration Leaderboard

Comprehensive BRD with functional requirements, API specifications, data models, privacy rules, acceptance criteria, and QA test scenarios.

2

Feature modules

2

REST APIs

16

Document sections

10

QA scenarios

Document version: 1.0

Module: Customer Mobile — Dashboard

Status: Phase 1 Implemented

Generated: 24 June 2026

H2O Online — Customer Dashboard: Devices & Hydration Leaderboard

Document type: Business Requirements Document (BRD)

Product module: Customer Mobile App — Dashboard

Feature set: Connected Devices (Coming Soon) + Monthly Hydration Leaderboard

Version: 1.0

Status: Implemented (Phase 1)

Audience: Product owners, business analysts, engineering, QA, stakeholders

1. Executive Summary

This document defines the business requirements, functional specifications, API contracts, data models, privacy rules, and acceptance criteria for two customer-dashboard capabilities within the H2O Online mobile application:

- My Devices section** — Presents planned wearable and smart-home integrations (Apple Watch, Smart Bottle, TV Hub) with honest **Coming soon** labelling instead of misleading "Connected" status.
- Monthly Hydration Leaderboard** — A gamified engagement feature that ranks customers by **total water intake (liters)** within a calendar month, with explicit **opt-in** controls for public listing and optional **monthly report** subscription.

These features support the platform's strategic goal of **health engagement beyond ordering** (Business Objective BO-06) while maintaining user trust and privacy through consent-based visibility.

2. Business Context & Objectives

2.1 Problem statement

Issue	Impact
Device tiles showed "Connected" with green status indicators	Users believed integrations were live; creates support burden and trust erosion
No competitive/social hydration motivation	Lower daily engagement with water-intake logging
No user control over public hydration ranking	Privacy risk if rankings were shown without consent

2.2 Business objectives

ID	Objective	KPI / Success measure
BRD-DL-01	Set accurate expectations for device integrations	Zero support tickets claiming "device already connected" after release
BRD-DL-02	Increase water-intake logging frequency	+15% weekly active loggers among opted-in customers (target)
BRD-DL-03	Drive return visits to customer dashboard	Leaderboard screen views ≥ 20% of MAU (target)
BRD-DL-04	Protect user privacy by default	100% of public leaderboard entries require explicit opt-in
BRD-DL-05	Prepare foundation for future wearable sync	Device catalogue visible; roadmap communicated as Coming soon

2.3 Stakeholders

Role	Interest
Customer (end user)	Accurate device status; optional social ranking; control over visibility
Product / Marketing	Engagement, retention, future device partnership narrative
Engineering	Clear API contracts, scalable monthly aggregation
Legal / Privacy	Consent for public display; name masking on leaderboard

Role	Interest
Operations / Support	Documented behaviour for FAQ and troubleshooting

3. Scope

3.1 In scope (Phase 1 — current release)

- Customer dashboard device carousel with **Coming soon** status for three device types
- Leaderboard entry point on dashboard (Monthly Hydration Leaderboard card)
- Dedicated **Leaderboard** screen (/leaderboard)
- User preference toggles: **Join leaderboard**, **Monthly report**
- Backend APIs for monthly rankings and preference updates
- Ranking algorithm based on summed volumeM1 from Water Intake records for calendar month
- Top 50 public entries returned; current user's rank computed when opted in
- Partial name display on leaderboard (first name + last initial)

3.2 Out of scope (Phase 1)

- Actual Apple Watch / HealthKit / Google Fit / smart bottle / TV hub pairing
- Push notifications or email delivery for monthly reports (preference stored only)
- Historical month picker UI (API supports year / month query params; UI defaults to current month)
- Admin portal management of leaderboard
- Rewards/points tied to leaderboard rank (reward points on dashboard remain separate/display-only)
- Society, corporate, or team-based leaderboards

3.3 Future phases (roadmap)

Phase	Capability
Phase 2	Email/push monthly hydration report for users with leaderboardMonthlyReport: true
Phase 3	Apple Watch / wearables OAuth and automatic intake sync
Phase 4	Smart bottle BLE pairing and real-time sync
Phase 5	TV Hub dashboard widget for household hydration summary
Phase 6	Month selector, badges, streak bonuses, friend invites

4. User Personas & Journeys

4.1 Primary persona: Health-conscious customer

Profile: Logs water intake 3–5 times per week; uses dashboard for hydration summary and orders.

Journey A — View devices

1. Opens Customer Dashboard
2. Scrolls to **My devices** section
3. Sees Apple Watch, Smart Bottle, TV Hub with **Coming soon** badge
4. Understands integrations are planned, not active

Journey B — Join leaderboard

1. Taps **Monthly Hydration Leaderboard** card on dashboard
2. Opens Leaderboard screen
3. Enables **Join leaderboard** toggle
4. Views personal rank and monthly liters after logging intake
5. Optionally enables **Monthly report** for future summaries

- 1. Opens Leaderboard screen to view top performers
- 2. Leaves **Join leaderboard** off
- 3. Sees informational banner: not listed in rankings
- 4. Can still browse public rankings of opted-in users

5. Functional Requirements — My Devices

5.1 Device catalogue

Device ID	Display name	Icon	Status (Phase 1)
1	Apple Watch	watch-outline	Coming soon
2	Smart Bottle	water-outline	Coming soon
3	TV Hub	tv-outline	Coming soon

5.2 UI requirements

Req ID	Requirement	Priority
DEV-UI-01	Section eyebrow: Connected ; section header: My devices	Must
DEV-UI-02	Horizontal scroll carousel of device cards	Must
DEV-UI-03	Each card shows device icon (gradient circle), name, and status text Coming soon	Must
DEV-UI-04	Amber Soon badge replaces green connected dot	Must
DEV-UI-05	Status text uses muted colour (not success green)	Must
DEV-UI-06	Device cards are non-interactive in Phase 1 (no false connect flow)	Should

5.3 Business rules

Rule ID	Rule
DEV-BR-01	No device may display "Connected" until backend pairing exists
DEV-BR-02	Copy must not imply data is syncing from wearables
DEV-BR-03	Create Profile "Connect device" toggle remains separate onboarding preference for future use

5.4 Implementation reference

Layer	File
Mobile UI	mobile/src/screens/DashboardScreen.js
Route	mobile/app/dashboard.tsx

6. Functional Requirements — Monthly Hydration Leaderboard

6.1 Dashboard entry point

Req ID	Requirement	Priority
LB-UI-01	Dashboard card titled Monthly Hydration Leaderboard	Must
LB-UI-02	Subtitle: ranked by water intake; tap to view and manage	Must
LB-UI-03	Trophy icon and chevron; navigates to /leaderboard	Must
LB-UI-04	Replaces prior placeholder "Summer Hydration Challenge" copy	Must

6.2 Leaderboard screen

Req ID	Requirement	Priority
LB-UI-10	Header: Hydration leaderboard with subtitle explaining monthly water-intake ranking	Must
LB-UI-11	Your preferences card with two switches	Must
LB-UI-12	Toggle 1: Join leaderboard — include my water intake in monthly rankings	Must
LB-UI-13	Toggle 2: Monthly report — receive monthly summary of hydration rank	Must
LB-UI-14	When opted out: banner explaining opt-in needed to appear in rankings	Must
LB-UI-15	When opted in: stats card showing rank (#N or —) and total liters for period	Must
LB-UI-16	Rankings list: rank badge, avatar/initials, masked name, liters	Must
LB-UI-17	Current user row highlighted; append (You) to name	Must
LB-UI-18	Top 3 ranks use trophy/medal styling	Should
LB-UI-19	Pull-to-refresh reloads leaderboard data	Must
LB-UI-20	Empty state when no opted-in users have intake data	Must
LB-UI-21	Unauthenticated users redirected to login on toggle interaction	Should

6.3 Preference business rules

Rule ID	Rule
LB-BR-01	<code>leaderboardOptIn</code> defaults to false for all users
LB-BR-02	<code>leaderboardMonthlyReport</code> defaults to false for all users
LB-BR-03	Only users with <code>role: customer</code> AND <code>leaderboardOptIn: true</code> appear in public rankings
LB-BR-04	Opting out removes user from subsequent ranking queries; historical rank not retained in UI
LB-BR-05	Monthly report toggle is independent of opt-in (user may want report without public listing — stored for Phase 2)
LB-BR-06	Ranking metric is total milliliters logged in period, displayed as liters (1 decimal)

6.4 Ranking algorithm

Period: Calendar month (default: current month in server local date logic via ISO date strings).

Steps:

1. Select all customers where `leaderboardOptIn === true`
2. Query `waterIntake` documents where `date` is between `YYYY-MM-01` and last day of month (inclusive)
3. For each document, sum `entries[].volumeMl`
4. Aggregate by `userId` ; sort descending by total ml
5. Assign rank 1..N (ties: same total → order by aggregation sort; no fractional ranks)
6. Return top **50** entries to client
7. Compute requesting user's rank even if outside top 50 (when opted in)

Display name masking: "Rohit Kumar" → "Rohit K."; single name unchanged.

Tie-breaking: Users with equal liters receive consecutive ranks based on sort order (standard competition ranking not required in Phase 1).

6.5 User stories

Story ID	As a...	I want to...	So that...
US-LB-01	Customer	See devices marked Coming soon	I am not misled about integrations
US-LB-02	Customer	Open a leaderboard from my dashboard	I can compare my hydration with others
US-LB-03	Customer	Opt in to the leaderboard	My logged intake counts toward my rank
US-LB-04	Customer	Opt out of the leaderboard	My data is not shown publicly

Story ID	As a...	I want to...	So that...
US-LB-05	Customer	See my monthly rank and liters	I know how I am performing
US-LB-06	Customer	Enable monthly report	I can receive summaries when that feature launches
US-LB-07	Customer	View top hydrators this month	I stay motivated to log water
US-LB-08	Privacy-conscious user	Browse leaderboard without joining	I can explore without exposing my data

7. API Specification

Base URL: {API_HOST}/api

Authentication: Bearer JWT on all endpoints below (same as existing auth middleware).

Content-Type: application/json

7.1 GET /leaderboard

Retrieves monthly leaderboard rankings, user preferences, and current user's stats.

Query parameters

Parameter	Type	Required	Description
year	integer	No	Calendar year (default: current). Valid range: 2020–2100
month	integer	No	Calendar month 1–12 (default: current)

Example request

```
GET /api/leaderboard
Authorization: Bearer <token>
```

```
GET /api/leaderboard?year=2026&month=6
Authorization: Bearer <token>
```

Success response: 200 OK

```
{
  "period": {
    "from": "2026-06-01",
    "to": "2026-06-30",
    "label": "June 2026"
  },
  "rankings": [
    {
      "rank": 1,
      "userId": "665a1b2c3d4e5f6789012345",
      "name": "Rohit K.",
      "avatarUrl": "https://...",
      "totalLiters": 42.5,
      "isCurrentUser": false
    }
  ],
  "preferences": {
    "optIn": true,
    "monthlyReport": false
  },
  "me": {
    "rank": 12,
    "totalLiters": 18.3,
    "optedIn": true
  }
}
```

Response field definitions

Field	Type	Description	
period.from	string	ISO date start of month (YYYY-MM-DD)	
period.to	string	ISO date end of month	
period.label	string	Human-readable month label	
rankings	array	Up to 50 entries, sorted by rank ascending	
rankings[].rank	integer	1-based position	
rankings[].userId	string	MongoDB user id	
rankings[].name	string	Masked display name	
rankings[].avatarUrl	string	Profile image URL or empty string	
rankings[].totalLiters	number	Rounded to 1 decimal place	
rankings[].isCurrentUser	boolean	True if row is authenticated user	
preferences.optIn	boolean	User's leaderboardOptIn setting	
preferences.monthlyReport	boolean	User's leaderboardMonthlyReport setting	
me.rank	integer \	null	Null if user not opted in
me.totalLiters	number	User's monthly total (0 if none)	
me.optedIn	boolean	Same as preferences.optIn	

Error responses

Status	Body	Condition
401	{ "error": "..."}	Missing or invalid token
500	{ "error": "..."}	Server/database error

7.2 PUT /leaderboard/preferences

Updates the authenticated user's leaderboard preferences.

Request body

Field	Type	Required	Description
optIn	boolean	No	Set leaderboard opt-in status
monthlyReport	boolean	No	Set monthly report preference

At least one field should be sent. Omitted fields are unchanged.

Example request

```
PUT /api/leaderboard/preferences
Authorization: Bearer <token>
Content-Type: application/json

{
  "optIn": true,
  "monthlyReport": true
}
```

Success response: 200 OK

```
{
  "optIn": true,
  "monthlyReport": true
}
```

Error responses

Status	Body	Condition
401	Unauthorized	Invalid session
404	{ "error": "User not found" }	User deleted
500	Server error	Save failure

7.3 Related APIs — Water Intake (ranking data source)

Leaderboard rankings depend on water intake logged via existing endpoints.

GET /water-intake/summary

Used by dashboard for 7-day chart; leaderboard uses same underlying `waterIntake` collection with monthly aggregation.

Parameter	Description
from	Start date YYYY-MM-DD
to	End date YYYY-MM-DD

POST /water-intake

Creates intake entry (feeds leaderboard when within month).

Body: { "date", "type", "quantity", "volumeMl" }

type enum: glass, jar, bottle, total

GET /users/me

Returns user profile including `leaderboardOptIn` and `leaderboardMonthlyReport` when present on User document.

7.4 Mobile API client mapping

Client method	HTTP
api.leaderboard.get(year?, month?)	GET /api/leaderboard
api.leaderboard.updatePreferences(body)	PUT /api/leaderboard/preferences
api.waterIntake.add(body)	POST /api/water-intake
api.waterIntake.summary(from, to)	GET /api/water-intake/summary
api.users.me()	GET /api/users/me

Client file: mobile/src/api/client.js

8. Data Model

8.1 User schema extensions

Collection: users

Model file: backend/models/User.js

Field	Type	Default	Description
leaderboardOptIn	Boolean	false	User consents to public monthly ranking
leaderboardMonthlyReport	Boolean	false	User wants monthly rank summary (delivery Phase 2)

8.2 WaterIntake (existing)

Collection: waterintakes

Model file: backend/models/WaterIntake.js

Field	Type	Description
userId	ObjectId	Reference to User
date	String	YYYY-MM-DD
entries	Array	{ type, quantity, volumeMl, createdAt }

Index: { userId: 1, date: 1 } unique

Leaderboard aggregation: Sum all entries.volumeMl per userId where date in [monthStart, monthEnd] .

9. Privacy, Security & Compliance

Topic	Requirement
Opt-in default	Public listing off by default (GDPR/consent-aligned design)
Name display	Partial masking on public leaderboard
Email/phone	Never exposed on leaderboard API
Role filter	Only customer role eligible for rankings
Authentication	All leaderboard endpoints require valid JWT
Data minimization	Rankings API returns only necessary display fields
Monthly report	Preference stored; no PII sent until delivery channel implemented

10. Non-Functional Requirements

NFR ID	Category	Requirement
NFR-01	Performance	GET /leaderboard p95 < 800ms for ≤10k opted-in users
NFR-02	Scalability	Aggregation pipeline indexed on userId + date
NFR-03	Availability	Same as core API (no separate service)
NFR-04	Accessibility	Toggle labels and hints readable; sufficient touch targets
NFR-05	Localization	English only Phase 1; labels externalized for future i18n

11. Acceptance Criteria

11.1 Devices

- ☐ All three device cards show **Coming soon** text
- ☐ No green "Connected" indicator visible
- ☐ Amber **Soon** badge displayed on each card

11.2 Leaderboard — Dashboard

- ☐ Leaderboard card navigates to Leaderboard screen
- ☐ Card copy references monthly water intake ranking

11.3 Leaderboard — Preferences

- ☐ Join leaderboard toggle persists via API
- ☐ Monthly report toggle persists via API
- ☐ Opted-out user has `me.rank: null` in API response
- ☐ Opted-in user sees rank after logging intake

11.4 Leaderboard — Rankings

- ☐ Only opted-in customers appear in rankings array
- ☐ Sort order is descending by total liters
- ☐ Maximum 50 entries returned
- ☐ Current user row marked with `isCurrentUser: true`
- ☐ Names masked per displayName rules

11.5 Integration

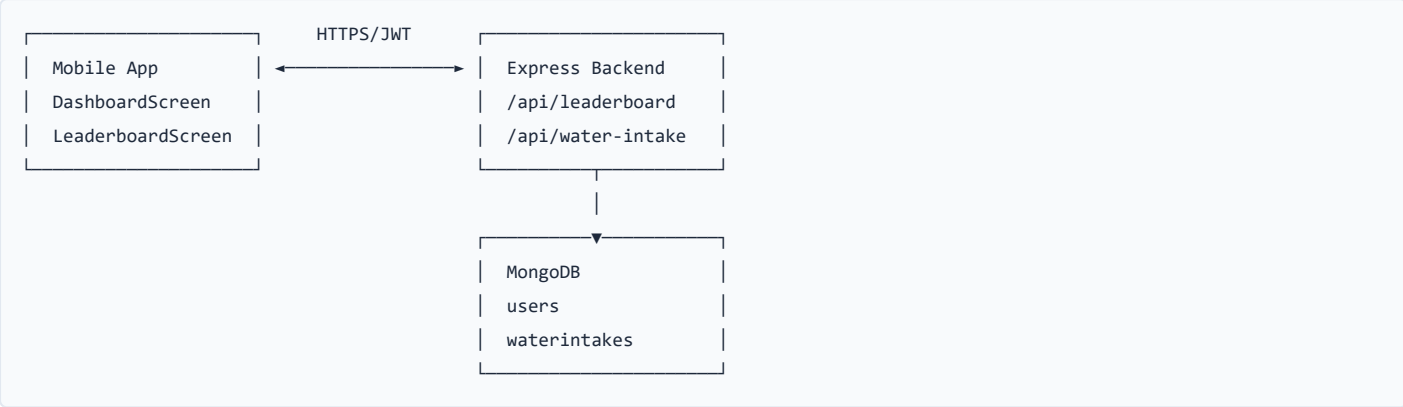
- ☐ New intake logged via POST /water-intake reflects in same-month rank after refresh
- ☐ Backend route mounted at `/api/leaderboard` in server.js

12. Test Scenarios (QA)

TC ID	Scenario	Expected result
TC-01	New user opens dashboard devices	All show Coming soon
TC-02	User opens leaderboard, toggles opt-in ON	API returns optIn true; user can appear after logging
TC-03	User toggles opt-in OFF	Removed from rankings on next fetch
TC-04	Two users, different intake totals	Higher liters = better (lower) rank number
TC-05	User opted in, zero intake	Rank computed; totalLiters 0
TC-06	No opted-in users	Empty rankings; empty state UI
TC-07	Invalid token on GET /leaderboard	401

TC ID	Scenario	Expected result
TC-08	Pull to refresh on leaderboard	Data reloads
TC-09	Query previous month via API	Correct period.from/to returned
TC-10	monthlyReport ON, optIn OFF	Both preferences saved independently

13. Technical Architecture



Backend route file: `backend/routes/leaderboard.js`

Server mount: `app.use("/api/leaderboard", require("./routes/leaderboard"))`

Mobile screens:

- `mobile/src/screens/DashboardScreen.js`
- `mobile/src/screens/LeaderboardScreen.js`
- `mobile/app/leaderboard.tsx`

14. Dependencies & Assumptions

Item	Assumption
Water intake logging	Users log via existing Water Intake screen
Auth	Customer logged in with JWT
Date storage	Intake dates stored as YYYY-MM-DD strings (UTC slice)
Monthly report	Phase 1 stores preference only; no notification job
Device integrations	Phase 1 is UI-only Coming soon state

15. Glossary

Term	Definition
Opt-in	User explicitly enables leaderboard participation
Monthly report	Future notification summarizing user's rank and intake
Total liters	Sum of logged volumeMI ÷ 1000, rounded to 1 decimal
Coming soon	Feature announced but not yet available for use
Masked name	First name plus last initial for privacy

16. Document Control

Version	Date	Author	Changes
1.0	June 2026	H2O Product Engineering	Initial BRD for Devices + Leaderboard Phase 1

End of document — H2O Online Customer Dashboard Devices & Hydration Leaderboard BRD v1.0